

Smart Workspace Manager

Combined Folder Structure Documentation

Version	2.0
Date	18 June 2026
Environment	Python 3.12, venv, pip
Cloud strategy	Azure free-first for personal use and testing

One backend and one frontend from project start to Phase 2.

Phase 2 adds professional web and Azure deployment files.

Phase 1 Streamlit UI is moved to legacy storage, not deleted.

Prepared as an independent Python learning project.

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1. Purpose and Structure Principles

This document defines the single repository structure used from the beginning of Phase 1 through the end of Phase 2. There are no phase1/ and phase2/ top-level folders. The project always has one backend/ and one frontend/. New professional web and cloud deployment folders are added only when Phase 2 begins.

Most important rule: The user interface calls backend services. It must never contain the main file-management, analysis, report, ML, or database logic. This is what allows Streamlit to be replaced by React without rewriting the system.

2. Lifecycle Legend

Marker	Meaning
[BOTH]	Created early and used in both phases.
[P1]	Actively used only by the Streamlit Phase 1 UI.
[P2+]	Added when Phase 2 starts.
[MOVED]	A Phase 1 item moved to legacy_streamlit/ after the React replacement is stable.
[FUTURE]	Not required in the two-phase plan; reserved for publication upgrades.

3. Full End-of-Phase-2 Folder Structure

```

Smart Workspace Manager/
|-- smart-workspace/           [BOTH] local venv; ignored by Git
|-- README.md                 [BOTH]
|-- requirements.txt           [BOTH] active Python backend packages
|-- requirements-dev.txt       [BOTH] tests and development tools
|-- requirements-streamlit.txt [P2+] legacy Streamlit dependencies
|-- pyproject.toml            [P2+] optional quality/tool config
|-- alembic.ini               [P2+] database migration config
|-- .gitignore                [BOTH]
|-- .env                      [BOTH] local secrets; ignored by Git
|-- .env.example              [BOTH]
|-- startup.sh                [P2+] Azure FastAPI startup command
|-- data/                     [BOTH] local development data root
|   |-- uploads/
|   |-- processed/
|   |   |-- csv/
|   |   |-- images/
|   |   |-- documents/
|   |   |-- pdf/
|   |   |-- others/
|   |-- reports/
|   |-- models/
|   |-- sample/
|-- docs/                     [BOTH]
|   |-- project_document.docx
|   |-- folder_structure.docx
|   |-- work_plan.docx
|   |-- api_documentation.md  [P2+]
|   |-- azure_deployment.md   [P2+]
|   |-- screenshots/
|-- backend/                  [BOTH]
|   |-- __init__.py
|   |-- main.py               [P2+] FastAPI entry point
|   |-- config/               [BOTH]
|   |   |-- __init__.py

```

```

|-- settings.py

-- database/ [BOTH]
|  |-- __init__.py
|  |-- db.py
|  |-- models.py
|  |-- repositories.py
|  |-- migrations/ [P2+] Alembic revisions
-- services/ [BOTH]
|  |-- __init__.py
|  |-- file_service.py
|  |-- automation_service.py
|  |-- storage_service.py
|  |-- analysis_service.py
|  |-- report_service.py
|  |-- ml_service.py
|  |-- auth_service.py [P2+]
-- schemas/ [P2+]
|  |-- __init__.py
|  |-- file_schema.py
|  |-- analysis_schema.py
|  |-- report_schema.py
|  |-- ml_schema.py
|  |-- user_schema.py
-- routes/ [P2+]
|  |-- __init__.py
|  |-- health_routes.py
|  |-- file_routes.py
|  |-- analysis_routes.py
|  |-- report_routes.py
|  |-- ml_routes.py
|  |-- auth_routes.py
-- middleware/ [P2+]
|  |-- __init__.py
|  |-- error_handler.py
-- dependencies/ [P2+]
|  |-- __init__.py
|  |-- database_dependency.py
|  |-- auth_dependency.py
-- utils/ [BOTH]
|  |-- __init__.py
|  |-- file_utils.py
|  |-- validators.py
|  |-- logging_config.py
|  |-- constants.py
-- frontend/ [BOTH]
|  |-- streamlit_app.py [P1 -> MOVED]
|  |-- pages/ [P1 -> MOVED]
|  |-- components/ [P1 -> MOVED]
|  |-- legacy_streamlit/ [P2+] created during transition
|  |  |-- streamlit_app.py
|  |  |-- pages/
|  |  |-- components/
|  |-- package.json [P2+]
|  |-- package-lock.json [P2+]
|  |-- vite.config.js [P2+]
|  |-- index.html [P2+]
|  |-- staticwebapp.config.json [P2+] Azure Static Web Apps config
-- src/ [P2+]
|  |-- main.jsx
|  |-- App.jsx
|  |-- pages/
|  |-- components/
|  |-- services/
|  |-- hooks/
|  |-- context/
|  |-- assets/
-- notebooks/ [BOTH]
|  |-- data_analysis_experiments.ipynb
|  |-- ml_experiments.ipynb
-- scripts/ [BOTH]

```

```

|-- create_folders.py
|-- generate_sample_data.py
|-- reset_local_database.py
|-- migrate_sqlite_to_azure_sql.py          [P2+ optional]
-- tests/                                  [BOTH]
  |-- unit/
  |-- integration/
  |-- api/                                [P2+]
  |-- conftest.py
  |-- test_smoke.py
-- infrastructure/                         [P2+]
  |-- azure/
    |-- README.md
    |-- app_service_settings.md
    |-- static_web_app_settings.md
    |-- sql_setup.md
-- .github/                                [P2+]
  |-- workflows/
    |-- deploy-backend.yml
    |-- deploy-frontend.yml

```

4. Files and Folders Created at Project Start

Item	Purpose	Lifecycle
smart-workspace/	Local Python 3.12 virtual environment.	Never committed.
README.md	Setup, run commands, features, limitations, and screenshots.	Expanded in both phases.
requirements.txt	Phase 1 active Python packages.	Updated in Phase 2 for the API backend.
requirements-dev.txt	pytest, coverage, formatting/linting tools if used.	Used in both phases.
.env / .env.example	Local secrets and documented environment variable names.	.env ignored; example committed.
data/	Local uploads, processed files, reports, models, and samples.	Used locally in both phases.
docs/	Project, structure, timeline, and screenshot documentation.	Expanded in Phase 2.
backend/config/	Application settings and path/database configuration.	Used in both phases.
backend/database/	Connection, models, and repository functions.	SQLite first; Azure SQL support later.
backend/services/	All reusable business logic.	Never removed or bypassed.
backend/utils/	Validation, paths, logging, and constants.	Used in both phases.
frontend/streamlit_app.py	Streamlit entry point.	Active only in Phase 1.
frontend/pages/	Streamlit multipage UI.	Active only in Phase 1.
frontend/components/	Reusable Streamlit UI helpers.	Active only in Phase 1.
notebooks/	Controlled experiments, not production logic.	Optional in both phases.
scripts/	Repeatable setup and maintenance scripts.	Expanded as needed.
tests/	Unit and integration tests.	Expanded with API tests in Phase 2.

5. Phase 1-Only Active UI Files

Phase 1 item	Purpose
frontend/streamlit_app.py	Configures the app, title, navigation, and shared page settings.
frontend/pages/1_Dashboard.py	Shows summary counts, recent activity, and quick links.
frontend/pages/2_File_Upload.py	Collects uploads and calls file_service.
frontend/pages/3_File_Manager.py	Lists, filters, downloads, renames, and deletes files through services.
frontend/pages/4_CSV_Analyzer.py	Collects analysis/cleaning choices and displays service results.
frontend/pages/5_Reports.py	Lists and downloads generated reports.
frontend/pages/6_ML_Demo.py	Runs the controlled ML workflow.
frontend/components/	Streamlit-specific chart, table, status, and navigation helpers.

These files should contain display and input logic only. They import backend services. They must not directly implement file movement, pandas cleaning, database CRUD, or model training.

6. Files and Folders Added in Phase 2

Phase 2 addition	Purpose
backend/main.py	Creates the FastAPI app, registers middleware/routes, and exposes the health endpoint.
backend/routes/	Maps HTTP requests to existing services.
backend/schemas/	Pydantic request and response models.
backend/middleware/	Central error handling, logging, and CORS-related processing.
backend/dependencies/	Database session and authentication dependencies.
backend/services/auth_service.py	Password hashing, token logic, and authentication operations.
backend/database/migrations/	Alembic migration revisions.
alembic.ini	Migration configuration.
startup.sh	Azure App Service startup command for FastAPI.
frontend/src/	React pages, components, hooks, context, and API client code.
frontend/package.json	React/Vite dependencies and scripts.
frontend/vite.config.js	Vite development and build configuration.
frontend/staticwebapp.config.json	Azure Static Web Apps routing/security configuration.
frontend/legacy_streamlit/	Preserves the completed Phase 1 UI after React becomes stable.
requirements-streamlit.txt	Allows the legacy Streamlit app to remain runnable without polluting the active API dependency set.
infrastructure/azure/	Human-readable Azure setup and deployment notes.
.github/workflows/	Optional GitHub Actions workflows for frontend and backend deployment.
tests/api/	FastAPI endpoint tests.
docs/api_documentation.md	Endpoint contracts, examples, errors, and authentication notes.
docs/azure_deployment.md	Free-tier resource setup, startup command, variables, limits, and smoke tests.

7. Files Moved, Retired, or Removed in Phase 2

Item	Action	Destination/replacement	Reason
frontend/streamlit_app.py	Move	frontend/legacy_streamlit/streamlit_app.py	React becomes the active frontend; retain the Phase 1 UI as a reference.
frontend/pages/	Move	frontend/legacy_streamlit/pages/	Streamlit page files are not used by React.
frontend/components/	Move	frontend/legacy_streamlit/components/	These helpers are Streamlit-specific.
streamlit in requirements.txt	Move dependency	requirements-streamlit.txt	The deployed FastAPI backend no longer needs Streamlit.
Local SQLite file in deployed app	Retire from cloud	Keep as local dev DB or migration backup	Azure SQL becomes the deployed Phase 2 database.
Phase 1 Streamlit startup command	Replace	FastAPI startup.sh	The App Service is repurposed from Streamlit to FastAPI.

Permanent deletion policy: No important source folder is deleted. backend/services/, backend/database/, backend/utils/, tests/, data/, docs/, notebooks/, and scripts/ remain. Phase 1 UI files are moved only after React passes the same feature tests.

8. Folder-by-Folder Responsibilities

Folder	Responsibility
backend/config	Reads environment variables and provides typed settings. It must not perform business operations.
backend/database	Creates sessions/connections, defines ORM models, and provides repository/data-access functions.
backend/services	Implements business use cases. It may call repositories and storage abstractions but not UI code.
backend/routes	Translates HTTP input to service calls and service results to HTTP responses.
backend/schemas	Defines external API contracts. Schemas are not ORM models.
backend/utils	Small general helpers with no application-state ownership.
frontend/legacy_streamlit	Frozen Phase 1 interface. Only bug fixes required to preserve reference behavior.
frontend/src/pages	Page-level React composition.
frontend/src/components	Reusable UI elements.
frontend/src/services	Axios/API functions only; no backend business logic.
frontend/src/context or hooks	Authentication state, shared UI state, and reusable frontend behavior.
data	Local development data only. Production-like cloud data must not be committed.
tests/unit	Fast, isolated function/service tests.
tests/integration	Database, storage, and service workflow tests.
tests/api	FastAPI request/response tests.
infrastructure/azure	Deployment instructions; no secrets.

9. Data, Database, and Azure Path Rules

Rule	Value
Local data root	./data
Azure free-test data root	/home/data
Database selector	DATABASE_URL environment variable
Local database URL	sqlite:///./data/smart_workspace.db
Cloud database	Azure SQL connection URL; value stored in App Service settings.
Uploaded file policy	Use server-generated names and retain original_name in the database.
Report/model policy	Use generated IDs or safe names; never trust a user path.
Git policy	Commit only .gitkeep/sample files, not uploads, databases, reports, models, or .env.
Future storage	Azure Blob Storage provider added only before publication.

10. Import and Dependency Rules

- frontend/ may import or call backend interfaces, but backend/ must never import frontend modules.
- backend/routes/ may import services, schemas, dependencies, and configuration.
- backend/services/ may import repositories, storage abstractions, utilities, and domain/ORM models as required.
- backend/database/ must not import Streamlit, React, or route modules.
- notebooks/ may import services for experiments, but production code must never depend on notebooks.
- scripts/ may call public service/database functions; duplicated business logic should be avoided.
- All filesystem roots and database URLs are resolved through settings rather than hard-coded absolute paths.

Phase 1:
Streamlit page → backend service → repository/storage → SQLite/local files

Phase 2:
React page → FastAPI route → backend service → repository/storage → Azure SQL/App Service files

11. Transition Procedure

1. Finish and tag the working Streamlit release.
2. Run service, database, and Streamlit manual acceptance tests.
3. Confirm that important logic is outside frontend/.
4. Add FastAPI main, schemas, routes, middleware, and dependencies.
5. Add React/Vite files under the existing frontend/ folder.
6. Connect React to a local FastAPI server.
7. Add Azure SQL support and Alembic migrations.
8. Deploy React to Static Web Apps Free and FastAPI to the existing App Service F1.
9. Run parity tests: upload, list, analyze, clean, report, ML, and login.
10. Move Streamlit files into frontend/legacy_streamlit/ only after parity is confirmed.
11. Move the Streamlit dependency into requirements-streamlit.txt.
12. Update README, API documentation, Azure documentation, and screenshots.

12. Creation Commands and .gitignore

12.1 Phase 1 PowerShell Folder Commands

```
mkdir data, docs, backend, frontend, notebooks, scripts, tests
mkdir data\uploads, data\processed, data\reports, data\models, data\sample
mkdir data\processed\csv, data\processed\images, data\processed\documents
mkdir data\processed\pdf, data\processed\others
mkdir backend\config, backend\database, backend\services, backend\utils
mkdir frontend\pages, frontend\components
mkdir tests\unit, tests\integration
```

12.2 Phase 2 Addition Commands

```
mkdir backend\routes, backend\schemas, backend\middleware, backend\dependencies
mkdir backend\database\migrations
mkdir frontend\src, frontend\src\pages, frontend\src\components
mkdir frontend\src\services, frontend\src\hooks, frontend\src\context, frontend\src\assets
mkdir frontend\legacy_streamlit
mkdir tests\api
mkdir infrastructure, infrastructure\azure
mkdir .github, .github\workflows
```

12.3 Recommended .gitignore

```
# Python environments and caches
smart-workspace/
.venv/
__pycache__/_
*.py[cod]
.pytest_cache/
.coverage
htmlcov/

# Secrets
.env
*.pem
*.key

# Runtime data
data/uploads/*
data/processed/*
data/reports/*
data/models/*
*.db
*.sqlite
*.sqlite3

# Keep empty folders
!data/uploads/.gitkeep
!data/processed/.gitkeep
!data/reports/.gitkeep
!data/models/.gitkeep

# Jupyter
.ipynb_checkpoints/

# React/Vite
frontend/node_modules/
frontend/dist/

# OS/editor
.DS_Store
Thumbs.db
.vscode/settings.json
```

13. Final Checklist

Check	Result
One backend folder	Yes - backend/ is reused.
One frontend folder	Yes - frontend/ holds the active React UI and legacy Streamlit UI after transition.
No phase-based top-level folders	Yes.

Check	Result
Phase 1 business logic reusable	Required inside backend/services/.
Phase 2 additions identifiable	Marked [P2+] in this document.
Phase 1 UI preserved	Moved to frontend/legacy_streamlit/.
Cloud configuration separated	Environment variables and infrastructure/azure notes.
Free-first Azure supported	F1 backend, Static Web Apps Free, Azure SQL free offer.
Docker required	No.
Conda required	No.